

In this lab sheet do **not use the C++ array class** use standard C arrays. Do not use the STL. Make sure to reference all the sources you used to solve the problem.

Classes -Safe Array

Statement: As we know in C, there is no check to determine whether the array index is out of bounds. During program execution, an out-of-bound array index can cause serious problems. Also, recall that in C++ the array index starts at 0.

Task Implementing and test (using a main program) a class called `safeArray` that solves the out-of-bound array index problem and allows the user to access the array contents via an `at` method which takes the index position of the desired element in the array. The signature for this methods is `int at(int index)` Every object of type `safeArray` should be an array of type `int`. During execution, when accessing an array component, if the index is out of bounds, the program must terminate with the error message "array out of bounds" and return -1.

Also implement a constructor that makes a `safeArray` object with a default size of 10 integers. You should then overload the constructor to allow the user to specify the `safeArray` size. The signatures for these methods are `safeArray()`; and `safeArray(int size)`;

Then add a destructor that frees up any memory used and sets the pointer to NULL. The signatures for the destructor is `~safeArray()`;

Usage example main.cpp,

```
#include<iostream>
Using namespace std;
#include"safearray.h"

safeArray mySafeArray; //default safeArray of size 10
safeArray anotherSafeArray(9); //a safeArray of size 9
int element = 0;
cout << (element = mySafeArray.at(2)) << endl; //
```

You must separate class declaration from definition and for this lab you will submit the following 3 files:
`safearray.h` `safearray.cpp` and `main.cpp`